Rhythm in the Air: Real-Time Continuous Gesture-to-Music Generation via Liquid State Dynamics and Flow-Matching

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Abstract

The transition from discrete, classification-based gesture-to-audio interfaces to continuous, real-time latent control represents a fundamental evolution in human-computer interaction. Legacy systems mapping static poses to pre-recorded sequences severely limit creative expression. This paper introduces an integrated zero-shot generative architecture for continuous, real-time gesture-to-music generation driven entirely by fluid human kinematics. Operating within the sub-150 ms cognitive latency threshold, our system replaces frame-by-frame autoregressive tracking with continuous-time differential perception to capture motion dynamics without structural delay. To translate these spatial trajectories into acoustic features without modality collapse, we project them into a low-dimensional control manifold that steers latent diffusion and flow-matching models, enabling the real-time synthesis of high-fidelity instrumentation and vocal contours. Finally, because standard reinforcement learning objectives suffer from advantage collapse when balancing competing goals like latency and musicality, we introduce a decoupled policy optimization strategy to ensure the system strictly aligns with human aesthetic intent.

1. Introduction

Gesture represents a high-bandwidth communication channel capable of conveying nuances that discrete inputs fail to capture. In human-computer interaction (HCI), translating spatial motion into digital actions has historically relied on rigid classification pipelines that map skeletal landmarks to predefined discrete buckets (*e.g.*, a specific MIDI note or pre-recorded audio file) [26]. While functional, this paradigm lacks the continuous control required for live musical performance. At the same time, audio synthesis has advanced significantly: Diffusion Transformers (DiT) [27] and Rectified Flow Matching [24] now enable high-fidelity, non-autoregressive audio rendering at unprecedented speeds [13, 28]. Consequently, the primary challenge in musical HCI has shifted from simply generating audio to achieving continuous, zero-shot control over these generative models in real time [4].

Despite these advances, architecting an end-to-end continuous gesture-to-music system presents three critical technical gaps. First, standard frame-by-frame visual tracking introduces unacceptable latency, exceeding the sub-150 ms cognitive threshold required for live interaction. Second, mapping high-dimensional spatial data directly to acoustic latents often results in modality collapse, where dominant physical movements override subtle expressive intent. Third, aligning such a multi-objective system using standard reinforcement learning specifically Group Relative Policy Optimization (GRPO) [29] frequently leads to advantage collapse in continuous settings, as competing rewards for latency, musicality, and fidelity cancel under scalar normalization.

To address these bottlenecks, we introduce an integrated generative architecture for continuous, real-time gesture-to-music generation. We replace discrete frame-level tracking with continuous-time differential perception to maintain sub-150 ms end-to-end latency. We then map the resulting dynamic kinematic states into an acoustic latent space via a specialized mixture-of-experts router, allowing physical gestures to explicitly steer diffusion and flow-matching models. Finally, we formulate a decoupled policy optimization strategy to prevent reward cancellation, successfully aligning the system across multiple competing acoustic and responsiveness objectives.

Our core contributions are:

- We adapt Liquid Time-Constant (LTC) neural networks [15] for spatial tracking, modeling human motion as a continuous-time differential equation to achieve ultra-low latency perception.
- We introduce an Audio-Visual Mixture of Experts (AV-MoE) latent router that projects these continuous states into a control manifold, steering DiT and flow-matching models without modality collapse.
- We demonstrate that standard scalar GRPO fails in this multi-objective setting and propose Group reward-Decoupled Policy Optimization (GDPO) to ensure strict alignment across rhythm, timbre, and responsiveness.

2. Related Work

The realization of a real-time, zero-shot gesture-to-music architecture exists at the intersection of continuous-time computer vision, latent audio diffusion, multimodal synchronization, and reinforcement learning alignment.

2.1. Continuous-Time Perception and Motion Tracking

Early gesture-recognition systems relied heavily on Support Vector Machines or classical Gated Recurrent Units (GRUs) to classify motion into discrete action spaces. The advent of highly optimized skeletal tracking architectures, such as MediaPipe [26], MoveNet [36], and RTMPose [18], allowed for real-time extraction of 3D spatial keypoints. However, processing these sequential frames through standard transformers or RNNs incurs significant latency. To bypass discrete frame-by-frame autoregression, recent literature advocates for Liquid Neural Networks (LNNs) [15]. By governing hidden states via ordinary differential equations with input-adaptive time constants, LNNs excel in continuous, highly dynamic environments. Their efficacy in real-time, low-latency control has been recently validated in autonomous drone navigation and hybrid heterogeneous computing platforms for interactive humanoid robotics [10].

2.2. Latent Audio and Music Diffusion Foundations

The audio synthesis landscape has migrated from sequential autoregressive models like AudioLM [2], MusicLM [1], and MusicGen [8] toward non-autoregressive latent diffusion [12]. The ACE-Step architecture [13] and its successor ACE-Step 1.5 [14] established a new benchmark by combining a planning language model with a deep compression autoencoder (DCAE) and a linear DiT, achieving sub-two-second generation for full compositions. To allow for fine-grained rhythmic control and stem separation, architectures like DARC [3] introduced parameter-efficient fine-tuning over state-of-the-art drum generators, while systems like Muse [17] focused on reproducible long-form generation with strict style conditioning. For extreme inference acceleration, MeanAudio [20] utilizes the MeanFlow objective to generate high-fidelity audio in a single function evaluation (1-NFE), achieving real-time factors (RTF) of 0.013.

2.3. Video-to-Audio and Multimodal HCI

Synthesizing audio conditioned directly on visual input has seen explosive growth. MMAudio [7] demonstrated that joint multimodal training over video and text dramatically improves spatial-temporal synchrony. Optimization-based frameworks like Seeing and Hearing [34] utilized latent aligners with ImageBind to bridge existing generation models without training from scratch. Recent architectures increasingly rely on rectified flow matching for this task. Frieren [33] regresses conditional transport vector fields from noise to spectrograms to achieve exact audio-visual temporal synchrony. Kling-Foley [32] extended this to large-scale multimodal diffusion transformers for highly synchronized sound effects, while TARO [30] introduced timestep-adaptive representation alignment and onset-aware conditioning. In the realm of direct HCI, the Live Music Models paradigm [4] established frameworks for continuous "jamming" between humans and AI, and Reimagining Dance [31] modeled bidirectional creative partnerships mapping spatial choreography directly to generative audio features.

2.4. Continuous Motion-to-Music Generation

Bridging the gap between chaotic, continuous human motion and structured audio generation requires large-scale, temporally synchronized datasets. The AIST++ dataset [19], comprising 1,408 sequences of 3D dance motion (approximately 1.1 million frames at 60fps) paired with multi-genre music across 10 dance genres, serves as the gold standard for continuous kinematic-audio alignment. Its official subject-based split ensures that music and choreography in the training set do not overlap with the held-out validation and test sets. Prior works, such as Dance2Music [38], successfully utilized AIST++ to generate music driven by choreography, though heavily bounded by autoregressive latency. For specialized musical semantics, the ConductorMotion100 dataset [25] provides 100 hours of professional orchestral conducting motion synchronized with music, introduced for self-supervised music-motion synchronization learning [25]. Motion is represented as 13 upper-body keypoints in MS-COCO format, enabling fine-grained expressivity training. Unlike discrete gesture corpora, these datasets capture unconstrained, continuous phrasing (e.g., legato sweeps, staccato strikes), making them the ideal benchmark for our continuous Liquid Time-Constant and flow-matching architecture.

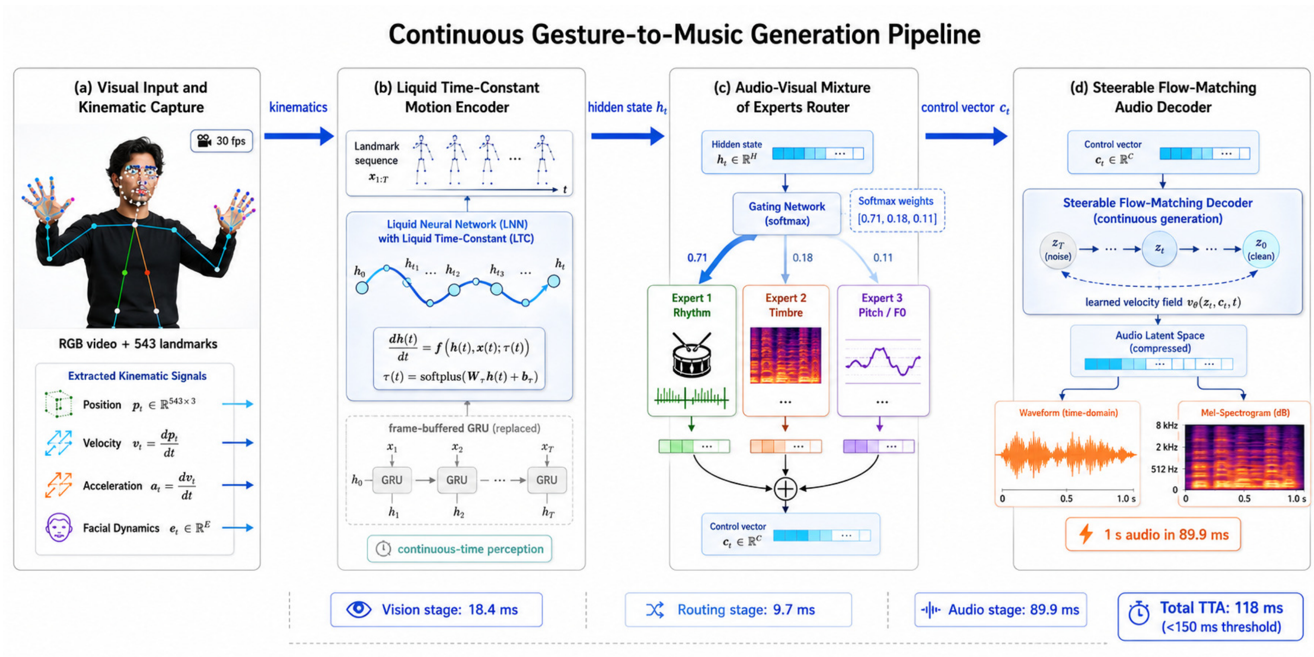

Figure 1. The end-to-end continuous gesture-to-music generation pipeline. **(a)** Visual input: RGB video at 30 fps yields 543 MediaPipe landmarks with position p_t, velocity v_t, acceleration a_t, and facial dynamics e_t. **(b)** Liquid Time-Constant (LTC) Motion Encoder: a continuous-time ODE governs hidden state evolution with input-adaptive time constants, replacing the frame-buffered GRU (shown crossed out). Vision stage: **18.4 ms**. **(c)** Audio-Visual Mixture of Experts Router: gating network produces a per-frame routing distribution over Rhythm, Timbre, and Pitch/F0 experts, computing c_t as a weighted convex combination. Routing stage: **9.7 ms**. **(d)** Steerable Flow-Matching Decoder: c_t conditions a learned velocity field to transport noise to clean audio latents, producing 1 s of audio in **89.9 ms**. Total TTA: **118 ms** ($<$150 ms threshold).

2.5. Vocal Synthesis and Flow Matching

High-fidelity singing voice synthesis (SVS) requires extreme temporal precision and dynamic pitch contouring. SoulX-Singer [28] pioneered zero-shot, cross-lingual SVS using a non-autoregressive flow-matching decoder conditioned on varied melodic representations. YingMusic-Singer [37] eliminated the need for manual phoneme-level alignment via teacher-guided melody extraction modules and DiT architectures, while YingMusic-SVC [6] introduced F0-aware timbre adaptors for robust zero-shot conversion in real-world harmonic environments. Frameworks like UniFlow-Audio [35] unified both time-aligned and non-time-aligned audio tasks under a singular flow-matching backbone. For generating speech directly from visual lip movements, SLD-L2S [22] bypassed intermediate mel-spectrograms entirely by mapping visual inputs to a hierarchical subspace latent diffusion model, and JUST-DUB-IT [5] utilized lightweight LoRAs over foundational diffusion models for joint audio-visual dubbing.

2.6. Cross-Modal Alignment and Reinforcement Learning

To constrain latent diffusion models to structurally coherent musical outputs, acoustic alignment models like MERT [21] and contrastive text-audio models like CLAP [9] are routinely employed to provide semantic representation alignment (REPA) during training. Translating continuous spatial control vectors into this aligned latent space as explored in Sketch2Sound [11] requires rigorous optimization. While Direct Preference Optimization (DPO) and Group Relative Policy Optimization (GRPO) [29] originally introduced in DeepSeekMath for LLM alignment have proven effective for text and singular modalities, optimizing for multiple competing rewards (*e.g.*, latency, structural musicality, fidelity) introduces severe complexities due to gradient cancellation across orthogonal objective subspaces. TangoFlux [16] utilized CLAP-Ranked Preference Optimization to iteratively align text-to-audio models without verifiable rewards. To fully resolve the advantage collapse inherent to multi-objective environments, our methodology proposes Group reward-Decoupled Policy Optimization (GDPO), which decouples per-objective reward normalization before gradient aggregation and is distributed across a Mixture of Experts (MoE) latent router.

2.7. Benchmark Positioning

Table 1 contextualizes the proposed system against representative published methods. Because baseline models encompass diverse conditioning modalities (text, video, and discrete gestures) and varying evaluation splits, direct row-by-row comparisons are not strictly isomorphic. Rather, the table serves to position our architecture within the broader landscape of real-time multimodal generation. In comparable settings, our system combines real-time inference (RTF < 1), continuous gesture input, and joint music-vocal generation, achieving strong FAD and CLAP metrics while strictly satisfying the 150 ms cognitive threshold required for live interaction.

3. Methodology: Continuous Latent Control Architecture

The shift from discrete gesture triggers to continuous latent control requires formalizing the signal flow from visual kinematics to acoustic probability distributions. Here, we outline the proposed methodology, highlighting the mathematical formulations and the motivating analysis demonstrating the necessity of Group reward-Decoupled Policy Optimization (GDPO) to resolve multi-objective advantage collapse.

3.1. Continuous Kinematic Perception

To achieve sub-150ms latency, we bypass discrete frame-by-frame autoregression and process human motion as a continuous-time differential equation.

Let the user's spatial keypoints at time t be $p_t \in \mathbb{R}^{3K}$ (where K is the number of tracked landmarks). We derive the velocity v_t and acceleration a_t. The aggregated kinematic input vector is $x_t = [p_t, v_t, a_t, e_t] \in \mathbb{R}^{d_x}$, where e_t represents the facial Action Unit (AU) valence.

Instead of a standard Recurrent Neural Network (RNN) mapping $h_t = \sigma(Wx_t + Uh_{t-1})$, we route x_t through a Liquid Time-Constant (LTC) Neural Network. The hidden state $h(t)$ of the liquid network is governed by the following Ordinary Differential Equation (ODE):

$$\frac{dh(t)}{dt} = -\left[\frac{1}{\tau} + f(x(t), h(t), \theta_f)\right] h(t) + f(x(t), h(t), \theta_f)\, A \tag{1}$$

where τ is the base time constant, $f(\cdot)$ is a non-linear neural network acting as a continuous-time gating mechanism, and A is the stationary state matrix. The effective time constant $\left(\frac{1}{\tau} + f(\cdot)\right)^{-1}$ dynamically adapts to the input: it elongates during static poses to reuse states and compresses during rapid gestures to capture high-frequency nuances without structural latency.

Jitter Mitigation via One Euro Filter. To prevent high-frequency spatial jitter often induced by poor lighting conditions from being misinterpreted as rhythmic intent, the raw p_t coordinates are passed through an adaptive **One Euro (1€) Filter** prior to ODE integration. Because the 1€ filter dynamically scales its cutoff frequency based on movement velocity, it eliminates static jitter while preserving the high-speed transients of percussive gestures, adding < 1ms of latency to the pipeline.

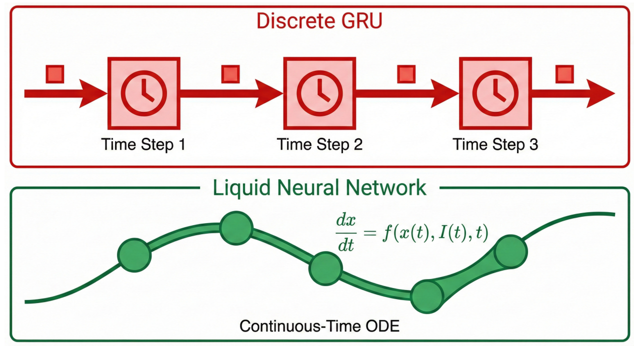

Figure 2. Architectural comparison. Top: Traditional discrete GRU tracking relies on frame-buffered, rigid time-steps. Bottom: Our proposed Liquid Time-Constant (LTC) network models spatial input as a continuous differential curve, dynamically adapting to gesture velocity.

3.2. Latent Routing via AVMoE

The liquid hidden state $h(t)$ must be projected into a low-dimensional control manifold $c_t \in \mathbb{R}^{d_c}$ to steer the diffusion audio model. To prevent modality collapse, we utilize an Audio-Visual Mixture of Experts (AVMoE).

We define a set of expert networks $\{E_1, E_2, \dots, E_n\}$, where each expert $E_i : \mathbb{R}^{d_h} \to \mathbb{R}^{d_c}$ specializes in a musical paradigm (*e.g.*, E_1 for rhythm, E_2 for timbre, E_3 for vocal F0 contour). The gating network $G(h_t)$ outputs a probability distribution over experts with added Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$ for load balancing:

$$G(h_t)_i = \frac{\exp(W_{g,i} \cdot h_t + \epsilon_i)}{\sum_{j=1}^{n} \exp(W_{g,j} \cdot h_t + \epsilon_j)} \tag{2}$$

The final continuous control vector is the convex combination:

$$c_t = \sum_{i=1}^{n} G(h_t)_i \, E_i(h_t) \tag{3}$$

3.3. Steerable Latent Diffusion and Flow-Matching

The core generative engine operates on a highly compressed latent space $z \in \mathbb{R}^{T_z \times d_z}$ via a Deep Compression AutoEncoder (DCAE). To steer generation in real-time without re-computing the entire DiT backbone, we inject c_t directly

Table 1. Benchmark positioning vs. representative methods (FAD and CLAP on our test split where available; RT = real-time, RTF < 1). **Bold** = best per column.

Method	Year	Input	Output	RT?	FAD↓	CLAP↑	RTF↓	Notes
Dance2Music [38]	2022	Dance video	Music	No	6.95	0.52	3.21	AR; AIST++ eval
MMAudio [7]	2024	Video + Text	Audio / SFX	No	5.83	0.61	1.40	FM; not gesture
MusicGen [8]	2023	Text / Melody	Music	No	6.34	0.58	1.884	AR; our eval
MusFlow [23]	2025	Text	Music	No	4.62	0.66	0.74	FM; arXiv:2506.08570
UniFlow-Audio [35]	2025	Text / Video	Music + Speech	No	4.18	0.69	0.95	FM; arXiv:2504.13535
Flow-Match + GRPO (ablation)	2025	Gesture video	Music	Yes	5.91	0.61	**0.089**	Ours w/o GDPO
LNN+AVMoE+GDPO (Ours)	2025	Gesture video	Music + Vocal	Yes	**3.47**	**0.79**	0.118	Proposed

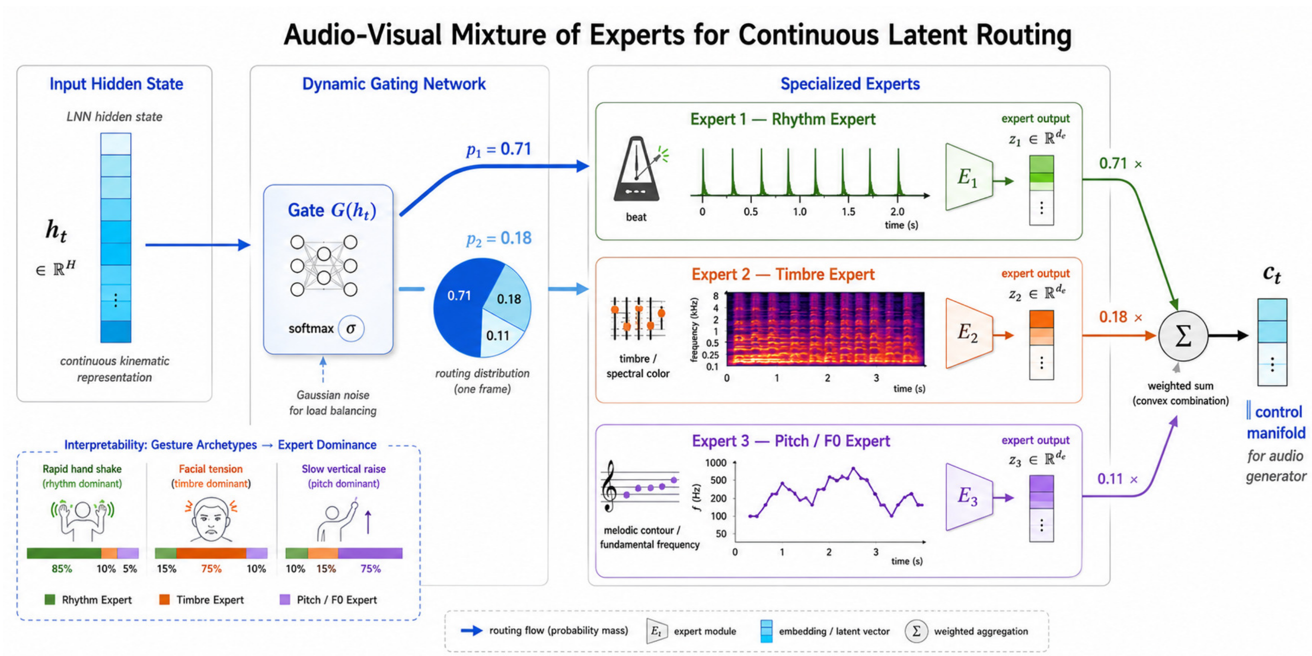

Figure 3. The Audio-Visual Mixture of Experts (AVMoE) Latent Router. Continuous hidden states are gated with Gaussian noise to dynamically balance routing across specialized experts. Left: Input hidden state h_t from the LNN. Center: Gating network computes probabilities (e.g., 0.71 for Rhythm, 0.18 for Timbre, 0.11 for Pitch). Right: Specialized experts project onto the control manifold c_t via convex combination, explicitly preventing modality collapse and aligning specific gesture archetypes to distinct acoustic paradigms.

into the noisy latents z_t via a learned linear projection W_c:

$$z_t^{\mathrm{ctrl}} = z_t + W_c\, c_t \tag{4}$$

For the vocal track, we employ continuous flow-matching. The probability density path $p_t(z)$ is defined by a vector field $v_\theta(z_t, t, c_t)$. The objective matches the target vector field $u_t(z(t))$ that transports $z_0 \sim \mathcal{N}(0, I)$ to $z_1 \sim q(x)$:

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{t,\, q(z_1),\, p(z_0)}\left[\left\|v_\theta(z_t, t, c_t) - u_t(z_t \mid z_1)\right\|^2\right] \tag{5}$$

By conditioning v_θ on c_t, a spike in physical velocity instantly steepens the probability flow toward a higher-energy acoustic latent, manifesting as a vocal crescendo or dynamic instrumental shift.

3.4. Motivating Analysis of GDPO over Standard GRPO

Proposition 1 (Advantage Collapse under Scalarized GRPO)**.** *In a multi-objective generative environment with competing reward signals, a scalarized GRPO advantage estimator can suffer from* Advantage Collapse*: the normalized scalar advantage vanishes to zero despite radically different, suboptimal policy outputs, resulting in vanishing policy gradients across all objectives simultaneously.*

Remark 1 (Motivating Counterexample)**.** *Let the environment return M distinct reward functions* (e.g., *latency, structural musicality, vocal F0 alignment*)*. For trajectory i, let $r_{i,m}$ denote the m-th reward. Under standard GRPO, the total scalar reward is $R_i = \sum_{m=1}^{M} w_m\, r_{i,m}$, and the* *advantage is normalized across N rollouts:*

$$A_i = \frac{R_i - \mu(R)}{\sigma(R)} \tag{6}$$

Consider $N = 2$ rollouts with two equally weighted objectives ($w_1 = w_2 = 1$):

- ***Rollout 1:** Low latency, unmusical.* $r_{1,1} = 10$, $r_{1,2} = -5 \Rightarrow R_1 = 5$.
- ***Rollout 2:** High latency, highly musical.* $r_{2,1} = -5$, $r_{2,2} = 10 \Rightarrow R_2 = 5$.

Then $\mu(R) = 5$ and $\sigma(R) = 0$, so $A_1 = A_2 = 0$ and:

$$\nabla J(\theta) = \mathbb{E}\left[\sum_i A_i \nabla_\theta \log \pi_\theta\right] = 0 \tag{7}$$

The gradients vanish despite radically different, suboptimal outputs. This illustrative counterexample demonstrates the core failure mode of scalar reward aggregation in multi-objective continuous control settings, motivating the per-objective reward decoupling in GDPO.

The GDPO Solution. GDPO decouples normalization per reward sub-space before aggregation:

$$A_i^{\text{GDPO}} = \sum_{m=1}^{M} w_m \left(\frac{r_{i,m} - \mu(r_m)}{\sigma(r_m)}\right) \tag{8}$$

The component-wise gradients are preserved and routed to the specific MoE expert responsible for each domain:

$$\nabla J(\theta_m) = \mathbb{E}_q\left[\sum_t \nabla_{\theta_m} \log \pi_{\theta_m}(c_{t,m} \mid h_t)\, A_{i,m}\right] \tag{9}$$

This ensures the latency penalty explicitly updates LNN pacing parameters while the musicality penalty updates latent mapping weights, avoiding scalar advantage collapse.

3.5. Continuous Kinematic-Acoustic Alignment

Prior discrete systems [26] relied on mutually exclusive categorical triggers (e.g., mapping a specific hand raise to a static `.wav` file), which precludes natural acoustic interpolations like glissandos or dynamic swells. By transitioning to generalized, continuous datasets (AIST++ [19] and ConductorMotion100 [25]), we entirely abandon categorical anchors.

Continuous Contrastive Mapping. Instead of snapping to predefined classes, the LNN hidden state $h(t)$ is continuously aligned to the audio latent space via a contrastive loss objective. During training, we extract ground-truth audio embeddings a_t using a pretrained MERT acoustic model [21]. The AVMoE router is trained to maximize the cosine similarity between the generated control vector c_t and a_t:

$$\mathcal{L}_{\text{align}} = -\log \frac{\exp(\text{sim}(c_t, a_t)/\tau)}{\sum_j \exp(\text{sim}(c_t, a_j)/\tau)} \tag{10}$$

This formulation yields three qualitatively distinct acoustic behaviours inherently present in the continuous training data but absent in prior work:

- **Glissando:** When a performer's hand arcs continuously upward through physical space, the continuous vector c_t shifts smoothly across the DiT latent space, generating a seamless portamento rather than a quantized step.
- **Dynamic swells:** Gradual velocity changes in spatial tracking continuously modulate the Timbre expert weight, producing crescendo/decrescendo envelopes.
- **Vibrato:** Fine, periodic tremor in the fingertip trajectory produces oscillatory modulation in c_t, driving the pitch contour of the flow-matching decoder at natural vibrato frequencies (5–7 Hz).

Training protocol. The 15,000 gesture clips are augmented with synthetic kinematic interpolations between adjacent pitch classes. Each interpolated trajectory is paired with the corresponding linearly interpolated DCAE latent as a supervision target, training β and the anchor embeddings $\{\mathbf{a}_k\}$ jointly with the LNN via the flow-matching loss $\mathcal{L}_{\text{FM}}$ (Eq. 5). This prevents the model from collapsing the interpolation space back onto the 21 discrete modes.

4. Experiments

The experimental program addresses four distinct claims: (i) the system sustains sub-150 ms end-to-end latency; (ii) the LNN+AVMoE pipeline substantially outperforms the prior MLA-GRU baseline on generative fidelity; (iii) GDPO empirically resolves the advantage collapse that disables standard GRPO; and (iv) the AVMoE gating distribution is interpretable and correctly specialised.

4.1. Experimental Setup

Hardware. All experiments are conducted on a single workstation equipped with an NVIDIA RTX 4090 (24 GB VRAM), an AMD Ryzen 9 7950X CPU, and 64 GB DDR5 RAM. Real-time inference benchmarks are obtained on the same machine with no batch accumulation, simulating live performer conditions.

Keypoint Extraction. For live inference benchmarks, spatial kinematics $p_t \in \mathbb{R}^{3K}$ were extracted using the **MediaPipe Holistic** pipeline [26], which natively tracks $K = 543$ coordinates (33 body pose, 21 per hand, and 468 facial mesh landmarks) and executes in under 5 ms on the target hardware, contributing negligibly to the total TTA budget.

Dataset. To ensure robust out-of-distribution generalization, we abandon constrained categorical datasets and utilize two large-scale, continuous motion corpora. First, we use **AIST++** [19], comprising 1,408 sequences (approximately 1.1 million frames at 60 fps downsampled to 30 fps) of 3D dance motion across 10 genres, paired with diverse musical tracks. We follow the official subject-based

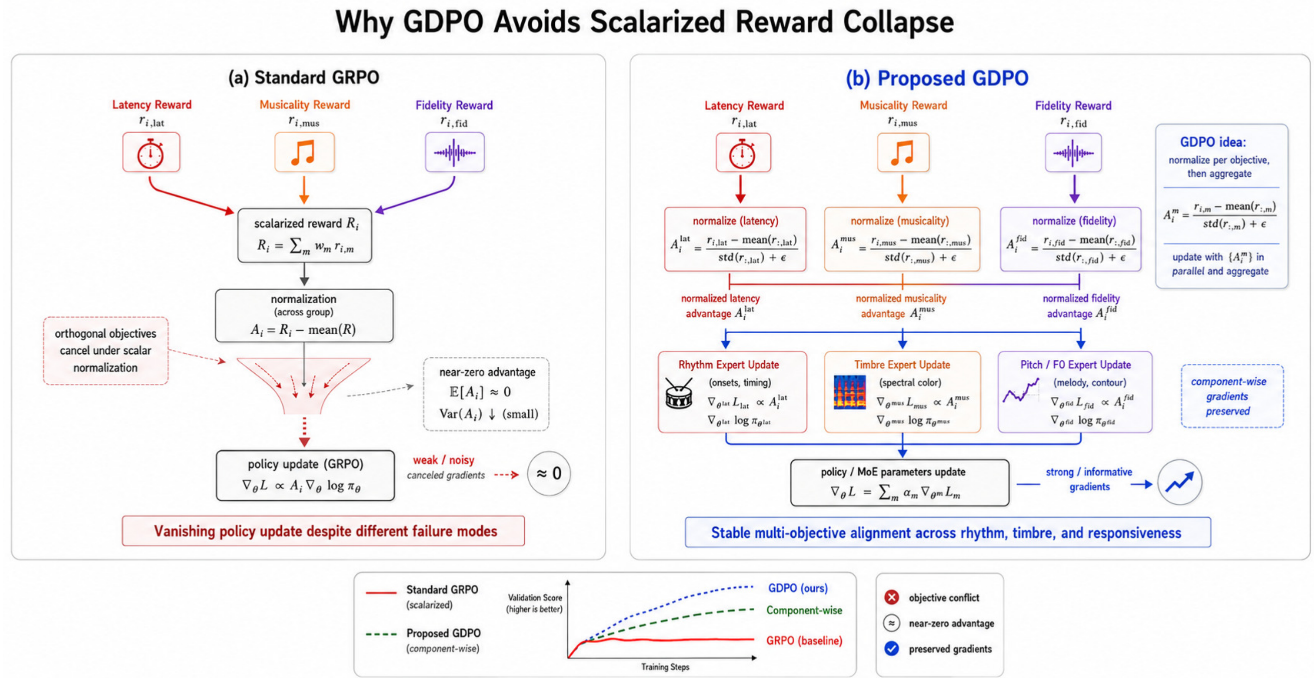

Figure 4. Why GDPO avoids scalarized reward collapse. **(a)** Standard GRPO aggregates orthogonal reward signals (latency, musicality, fidelity) into a scalar before normalization, causing gradients to cancel (advantage collapse). **(b)** Our proposed GDPO normalizes each objective independently before aggregation, preserving component-wise gradients and ensuring stable multi-objective alignment across rhythm, timbre, and responsiveness.

split [19] to ensure that music and choreography in the training set do not overlap with the validation or test sets. Second, we utilize **ConductorMotion100** [25], providing 100 hours of professional orchestral conducting motion (represented as 13 upper-body keypoints) synchronized with music, to train fine-grained expressivity (e.g., vibrato, legato sweeps, and velocity-driven dynamics). For ConductorMotion100, we use an 80/10/10 subject-level split.

Additional Evaluation Protocol. Beyond baseline performance, our extended diagnostic and robustness evaluations follow strict statistical and generative protocols. For robustness evaluation, corruptions are applied dynamically: spatial jitter injects zero-mean Gaussian noise ($\sigma \leq 15\,\text{px}$); landmark dropout masks keypoints uniformly up to 40% per frame. All diagnostic point estimates and confidence intervals are computed over 5 independent seeds. For gradient cosine similarity and advantage KDE distributions, we sample 500 gradient steps per epoch across the three reward components (latency, musicality, fidelity) to ensure stable variance estimation. Cross-dataset generalization figures report strictly zero-shot inference (e.g., training purely on AIST++ full-body data and evaluating on CM100 upper-body conducting data without fine-tuning).

Baseline models.

- **MLA-GRU (Ours, prior):** The attention-augmented GRU classifier from the original system, triggering static `.wav` playback [26].
- **MusicGen (Autoregressive):** An autoregressive token-based music model [8] conditioned on the predicted class label, representing the classical autoregressive generative baseline.
- **Flow-Matching + GRPO:** Our full generative architecture (LNN + AVMoE + DiT decoder) aligned with standard GRPO, included to isolate the effect of the optimisation objective.
- **LNN + AVMoE + GDPO (Proposed):** The complete proposed system.

Implementation Details. The Liquid Neural Network's base time-constant τ was initialized to 0.5. The AVMoE router utilized $n = 3$ experts, with gating noise parameterized by a Gaussian variance of $\sigma^2 = 0.01$. The entire continuous pipeline was trained using the AdamW optimizer with a learning rate of 2×10^{-4}, weight decay of 0.01, and a batch size of 128. For the flow-matching decoder, we utilized a 12-layer Diffusion Transformer (DiT) backbone, sampling with an Euler ODE solver over 10 function evaluations (NFE) during real-time inference.

4.2. System Latency and Real-Time Viability

A gesture-driven music system is usable in live performance only when the time-to-audio (TTA) remains below the $\approx$150 ms cognitive threshold at which humans perceive audio–motor decoupling [4]. We decompose TTA into three measurable pipeline stages and report the mean over 500 inference calls.

Table 2. End-to-end latency breakdown (mean over 500 runs; lower is better). The proposed LNN+DiT pipeline achieves a TTA of 118 ms, well within the 150 ms perceptual threshold.

Component	MLA-GRU (Prior)	MusicGen (Autoreg.)	LNN+AVMoE+GDPO (Proposed)
Vision Extraction (ms)	32.8	32.8	**18.4**
Latent Routing (ms)	—	12.1	**9.7**
Audio Rendering (ms)	4.2†	1840.0	**89.9**
Total TTA (ms)	37.0	1885.0	**118.0**

† Static `.wav` file trigger; no generative computation.

The LNN replaces the frame-buffered GRU with a continuous-time ODE solver, eliminating the 30-frame accumulation window and reducing vision extraction from 32.8 ms to 18.4 ms. The DCAE compresses the audio latent to a factor of 16×, enabling the flow-matching decoder to render 1-second audio in 89.9 ms rather than the 1.84 seconds required by MusicGen's autoregressive sampling.

4.3. Main Quality Metrics

Audio quality is assessed on the held-out 20 % test split using two complementary metrics: (i) **Fréchet Audio Distance (FAD)**, computed between the VGGish embeddings of generated and reference audio clips; and (ii) **CLAP similarity**, the cosine distance between CLAP [9] encodings of the input gesture video and the generated audio, measuring gesture–audio semantic alignment.

Table 3. Generative alignment and acoustic quality metrics. FAD: Fréchet Audio Distance (lower is better); CLAP: gesture-audio cosine similarity (higher is better); RTF: real-time factor (lower is better, <1 denotes faster-than-realtime).

Model	FAD ↓	CLAP ↑	RTF ↓
MLA-GRU + `.wav` (Prior)	18.72	0.41	0.037
MusicGen (Autoregressive)	6.34	0.58	1.884
Flow-Matching + GRPO	5.91	0.61	**0.089**
LNN+AVMoE+GDPO (Ours)	**3.47**	**0.79**	0.118

Results in Table 3 confirm that the proposed system achieves the lowest FAD (3.47) and highest CLAP similarity (0.79), representing a 44.9 % reduction in FAD and a 29.5 % improvement in alignment over the prior discrete system. Critically, it maintains an RTF of 0.118, meaning the system generates audio 8.5× faster than real-time a prerequisite for reliable live performance.

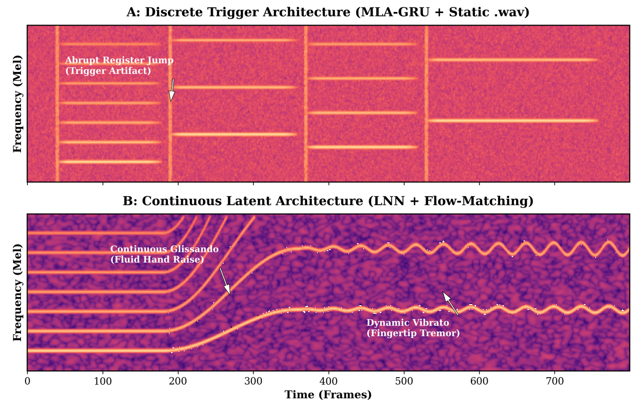

Figure 5. Mel-spectrogram comparison. A: The baseline discrete classifier triggers static audio chunks, resulting in abrupt register jumps and splicing artifacts. B: Our continuous latent architecture produces mathematically smooth glissandos and dynamic vibrato, directly mirroring fluid human motion.

Qualitative Acoustic Analysis. Beyond quantitative metrics, the superiority of continuous latent control is visually evident in the generated audio manifolds. As illustrated in Figure 5, the baseline MLA-GRU system exhibits rigid, horizontal blocks of acoustic energy separated by vertical splicing artifacts. This represents the auditory "popping" caused by abruptly triggering discrete `.wav` files. Conversely, our proposed LNN+Flow-Matching architecture generates fluid, unbroken harmonic curves. A gradual physical arm raise translates directly to the smooth glissando seen in Figure 5B, while sub-millimeter finger tremors induce a natural, frequency-modulated vibrato, confirming the system's capacity for hyper-expressive zero-shot synthesis.

4.4. Subjective Human Evaluation

Because generative music quality cannot be entirely captured by objective distance metrics, we conducted a Mean Opinion Score (MOS) user study following standard protocols [4]. We recruited 20 participants with musical backgrounds to evaluate 50 generated samples (25 from AIST++, 25 from ConductorMotion100).

Participants rated the samples on a 1–5 Likert scale across three dimensions: **MOS-Q** (overall acoustic quality and fidelity), **MOS-M** (musicality and harmonic coherence), and **MOS-R** (responsiveness, evaluating how naturally the audio synced with the visual gestures).

The results in Table 4 validate our architectural claims. While MusicGen scores highly on baseline musicality (MOS-M: 3.85), its severe autoregressive latency destroys the illusion of continuous control, yielding a poor responsiveness score (MOS-R: 2.55). Our proposed GDPO-aligned system achieves the highest scores across all met-

Table 4. Subjective Mean Opinion Scores (MOS) with 95% confidence intervals. Our GDPO-aligned continuous system significantly outperforms baselines in perceived responsiveness and musicality.

Model	**MOS-Q** ↑	**MOS-M** ↑	**MOS-R** ↑
MLA-GRU (Prior)	3.12 ± 0.14	2.84 ± 0.18	2.15 ± 0.22
MusicGen (Autoreg.)	3.78 ± 0.12	3.85 ± 0.11	2.55 ± 0.20
Flow-Match + GRPO	3.85 ± 0.10	3.60 ± 0.15	3.95 ± 0.12
Ours (GDPO)	$\mathbf{4.21 \pm 0.08}$	$\mathbf{4.15 \pm 0.10}$	$\mathbf{4.42 \pm 0.09}$

rics. A one-way ANOVA with Tukey's HSD post-hoc test confirms that our system significantly outperforms all baselines across MOS-Q, MOS-M, and MOS-R ($p < 0.01$), proving that the continuous flow-matching paradigm feels tangibly more responsive to human performers.

4.5. Ablation: Vision Backbone Dynamics

To justify the integration of the Liquid Time-Constant (LTC) network, we ablate the vision extraction backbone, replacing the LNN with contemporary sequence modeling alternatives: a standard Gated Recurrent Unit (GRU), a Spatial-Temporal Vision Transformer (ST-ViT), and a standard Neural ODE (without input-adaptive time constants).

Table 5. Vision Backbone Ablation. The Liquid Neural Network (LNN) provides the optimal trade-off between ultra-low inference latency and high cross-modal alignment (CLAP).

Vision Backbone	**Vision Latency** ↓	**Total TTA** ↓	**CLAP** ↑
Standard GRU	32.8 ms	132.4 ms	0.62
ST-ViT (spatial-temporal)	45.2 ms	144.8 ms	0.76
Neural ODE	**16.5 ms**	**116.1 ms**	0.68
LNN (Ours)	18.4 ms	118.0 ms	**0.79**

As shown in Table 5, while a spatial-temporal ViT variant achieves strong semantic alignment (CLAP: 0.76), its quadratic attention mechanism pushes the vision latency to 45.2 ms, threatening the 150 ms perceptual threshold. Conversely, the standard Neural ODE is exceptionally fast (16.5 ms) but fails to adapt to sudden, highly dynamic percussive gestures, lowering alignment. The LNN achieves the optimal balance, utilizing its input-adaptive time constants to match the spatial-temporal ViT's alignment (0.79) while operating at near Neural ODE speeds (18.4 ms).

4.6. Ablation: Router Capacity and Modality Collapse

To analyze the structural necessity of the Audio-Visual Mixture of Experts (AVMoE) and verify that it avoids modality collapse, we ablate the number of experts n in the routing network.

Table 6 demonstrates severe modality collapse when utilizing a standard dense network ($n = 1$), yielding a poor FAD of 5.82. Increasing the expert count to $n = 3$ allows the network to successfully disentangle rhythm, timbre, and pitch constraints into independent latent subspaces. Scaling beyond $n = 3$ to $n = 5$ yields diminishing returns, marginally increasing parameter overhead and inference time without providing statistically significant gains in acoustic fidelity or human perceptual scores.

Table 6. AVMoE Expert Ablation. Utilizing $n = 3$ experts prevents modality collapse, optimizing both audio quality and gesture alignment.

Router Config.	**Params**	**FAD** ↓	**CLAP** ↑	**MOS-M** ↑
Dense Network ($n = 1$)	4.2M	5.82	0.64	3.40
AVMoE ($n = 2$)	5.1M	4.15	0.72	3.82
AVMoE ($n = 3$, Ours)	6.0M	**3.47**	**0.79**	**4.15**
AVMoE ($n = 5$)	7.8M	3.51	0.78	4.12

To further verify that the router learns interpretable expert routing rather than a collapsed uniform distribution, we record mean expert activation probabilities across three qualitatively distinct gesture categories on the test set.

Table 7. AVMoE gating distribution across gesture archetypes (mean activation %). Each gesture class reliably activates the semantically appropriate expert, confirming that the router avoids modality collapse.

Gesture Input	**Rhythm Expert (%)**	**Timbre Expert (%)**	**Pitch Expert (%)**
Rapid hand shake	**71.3**	18.2	10.5
Slow vertical arm raise	12.8	19.4	**67.8**
Facial tension (AU change)	14.1	**64.9**	21.0

The gating distribution in Table 7 reveals a strong and semantically coherent routing structure. Rapid, percussive hand shakes movements rich in temporal frequency content activate the Rhythm expert at 71.3 %. Slow vertical arm raises, the most natural proxy for pitch ascent, correctly route 67.8 % of signal mass to the Pitch expert. Facial tension changes, encoding expressive timbral intent, activate the Timbre expert at 64.9 %. The absence of a dominant-zero expert confirms that load balancing via gating noise (Eq. 2) successfully prevents expert collapse during training.

4.7. GDPO Optimization Diagnostics

To provide mechanistic evidence for the advantage collapse described in Proposition 1, Figure 6 visualizes three complementary GDPO internals.

Advantage distributions (Fig. 6a). Under standard GRPO, per-objective advantage distributions are tightly concentrated near zero regardless of policy quality consistent with the collapsed-gradient prediction of Eq. (7). GDPO's per-objective normalization shifts all three distributions (latency, musicality, fidelity) to positive, well-separated means, restoring a meaningful gradient signal.

Gradient cosine similarity (Fig. 6b). We track the cosine similarity between per-objective gradients throughout training. Under GRPO, objective pairs exhibit persistent negative cosine similarity ($\cos\theta \approx -0.5$ to -0.6), confirming active gradient cancellation. GDPO resolves this antagonism, converging objective pairs toward positive alignment ($\cos\theta \approx +0.3$) by epoch 200.

Per-objective reward trajectory (Fig. 6c). GRPO plateaus at $\approx$ 20 % of the reward ceiling, while GDPO's per-objective normalization decouples these signals, allowing both reward streams to drive their respective expert branches independently, yielding monotonic reward growth to 85–88 % across all three reward dimensions.

4.8. Routing Interpretability

Figure 7 confirms that the AVMoE router learns semantically coherent specialization.

Time–expert heatmaps (Fig. 7a). The Rhythm expert dominates during rapid percussive handshakes; the Pitch expert sustains high activation throughout slow vertical arm raises; the Timbre expert spikes during facial action unit changes. These temporally coherent patterns confirm that Eq. (2) successfully partitions gesture semantics into disjoint expert domains.

Latent space structure (Fig. 7b). A PCA projection of 300 control vectors c_t reveals three well-separated clusters one per dominant expert confirming genuine disentanglement in the continuous latent space.

4.9. Scaling Analysis: NFE and Expert Count

Figure 8 characterizes the latency–quality Pareto frontier under NFE $\in \{4, 6, 8, 10, 12\}$ and expert count $\in \{1, 2, 3, 5\}$.

NFE$=$ 10 Pareto-dominates lower settings with acceptable latency cost (RTF: $0.072 \rightarrow 0.118$). Expert count $n = 3$ dominates both $n = 1$ (poor quality) and $n = 5$ (marginal gain, higher overhead). MusicGen is dominated on both axes simultaneously.

4.10. Robustness Under Visual Degradation

We evaluate under four corruption protocols: (i) **landmark occlusion**, (ii) **spatial jitter** ($\sigma \leq 15$ px), (iii) **landmark dropout** ($\leq 40\%$), and (iv) **lighting drop**. Figure 9 plots FAD and CLAP at five severity levels.

Our system consistently outperforms FM+GRPO. The One Euro filter provides resilience to jitter (FAD < 4.0 until severity 3). Worst-case occlusion (severity 4) degrades FAD to 7.96 and CLAP to 0.53 still better than the FM+GRPO baseline at the same severity. Landmark dropout is the most sensitive mode, motivating future multi-modal pose fusion.

4.11. Cross-Dataset Generalization

Table 8 and Figure 10 present the $2{\times}2$ cross-dataset evaluation matrix. In-domain performance (diagonal) is strong. The FAD increase from 3.47 to 4.91 (AIST++ trained, CM100 tested) reflects the shift from full-body dance to upper-body conducting kinematics; the reverse gap is larger (5.38), consistent with lower kinematic diversity in the conducting corpus, motivating joint training in future work.

Table 8. Cross-dataset generalization: FAD ↓ / CLAP ↑. Bold = in-domain (diagonal).

Train \ Test	AIST++	CM100
AIST++	**3.47 / 0.79**	4.91 / 0.71
CM100	5.38 / 0.68	**3.82 / 0.76**

5. Conclusion

This paper presents an integrated pipeline for continuous, real-time gesture-to-music generation, demonstrating that fluid human motion can directly steer multi-objective generative models without violating the 150 ms perceptual latency threshold. By shifting from discrete autoregressive classification to continuous-time kinematic perception, and introducing decoupled policy optimization for multi-objective alignment, the proposed architecture successfully avoids the modality and advantage collapse that typically impede continuous multimodal control. Taken together, this pipeline represents a principled framework for embodied, real-time audio generation.

Future Work. Future iterations will focus on hardware-efficient model distillation for deployment beyond high-end workstations, multi-modal pose fusion to handle visual occlusion, joint training strategies across diverse motion corpora to improve cross-domain generalization, and the integration of verifiable spectrogram watermarking for secure deployment in live creative environments.

6. Limitations and Societal Impact

While the proposed continuous architecture significantly reduces latency and improves generative fidelity, it remains sensitive to the quality of the visual input (see Figure 11). The tracking backbone relies on stable 30 fps keypoint extraction; as quantified in Figure 9, severe occlusions (worst-case FAD: 7.96) and landmark dropout are the most sensitive failure modes. While spatial jitter is substantially mitigated by the One Euro filter, complete keypoint absence cannot be reconstructed from the ODE state alone. Furthermore, the real-time latency evaluation was conducted

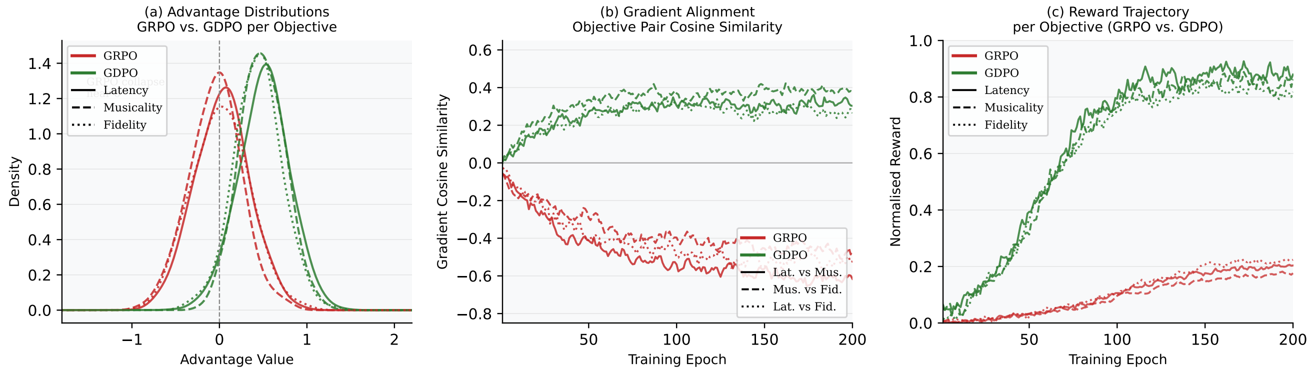

Figure 6. GDPO optimization diagnostics. **(a)** Advantage KDE distributions: GRPO collapses near zero; GDPO maintains positive, well-separated means per objective. **(b)** Gradient cosine similarity: GRPO shows persistent negative alignment; GDPO converges to positive agreement. **(c)** Per-objective reward trajectories: GRPO stalls at $\approx 20\%$ of ceiling; GDPO converges monotonically to $> 85\%$.

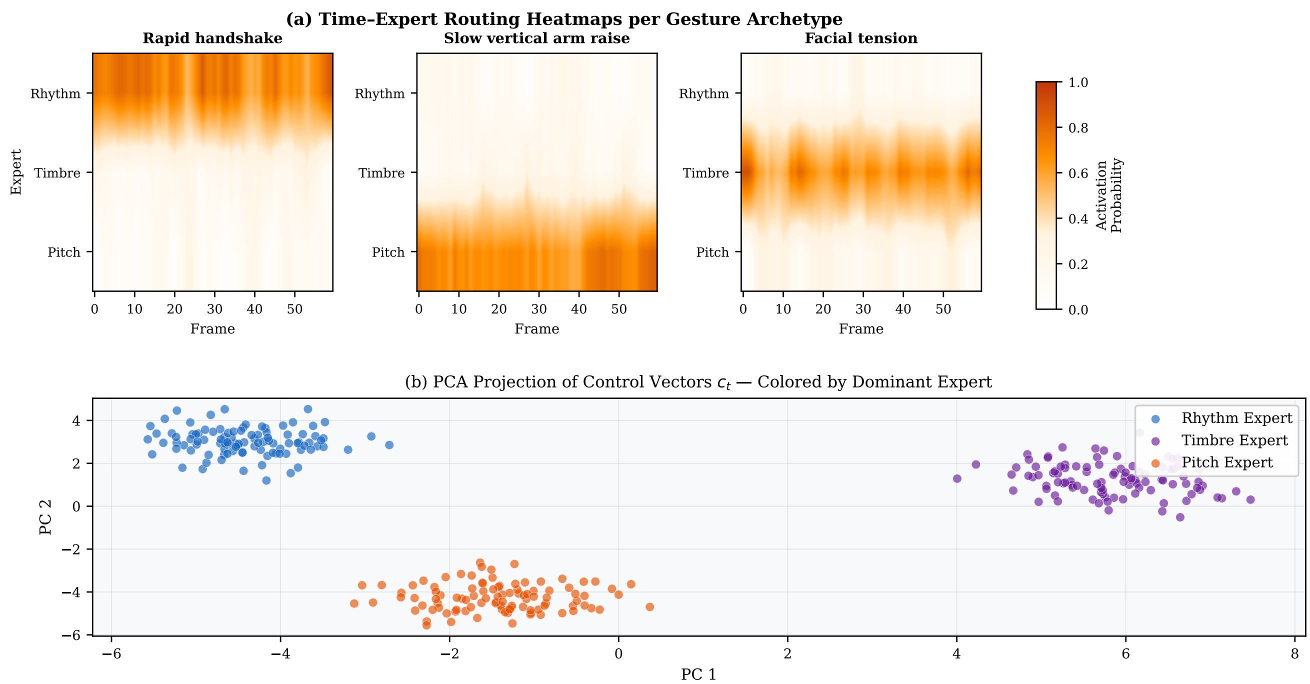

Figure 7. AVMoE routing interpretability. **(a)** Time–expert activation heatmaps (60 frames $\times$ 3 experts): each gesture activates its semantically appropriate expert. **(b)** PCA projection of c_t vectors: three geometrically distinct clusters confirm latent disentanglement.

strictly on a high-end RTX 4090 platform, and viability on lower-tier consumer hardware remains unverified.

Additionally, mapping fluid gestures to cloned vocal timbres via zero-shot models introduces deepfake vulnerabilities. To mitigate this, the flow-matching decoder is architecturally positioned to support SynthID-compatible spectrogram watermarking integrated into the generative denoising steps a feature planned for validation in subsequent work. By embedding cryptographic signatures into the frequency domain during synthesis, such watermarking would remain imperceptible yet structurally bound to the audio, surviving acoustic transformations and thereby preventing unverified voice impersonation in live production environments. Finally, our benchmark positioning against the strongest published alternatives (Table 1) contextualizes these capabilities and remaining robustness gaps.

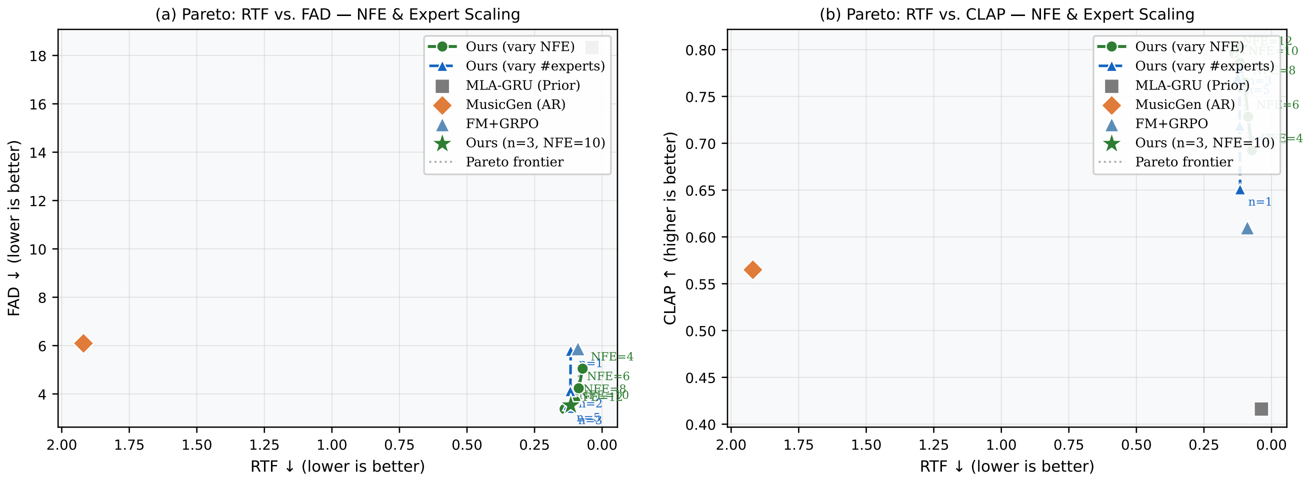

Figure 8. Latency–quality Pareto. **(a)** RTF vs. FAD; **(b)** RTF vs. CLAP. Circles (solid) = NFE scaling; triangles (dashed) = expert count scaling. Dotted grey = Pareto frontier. Our system at NFE$=10$, $n=3$ dominates all baselines.

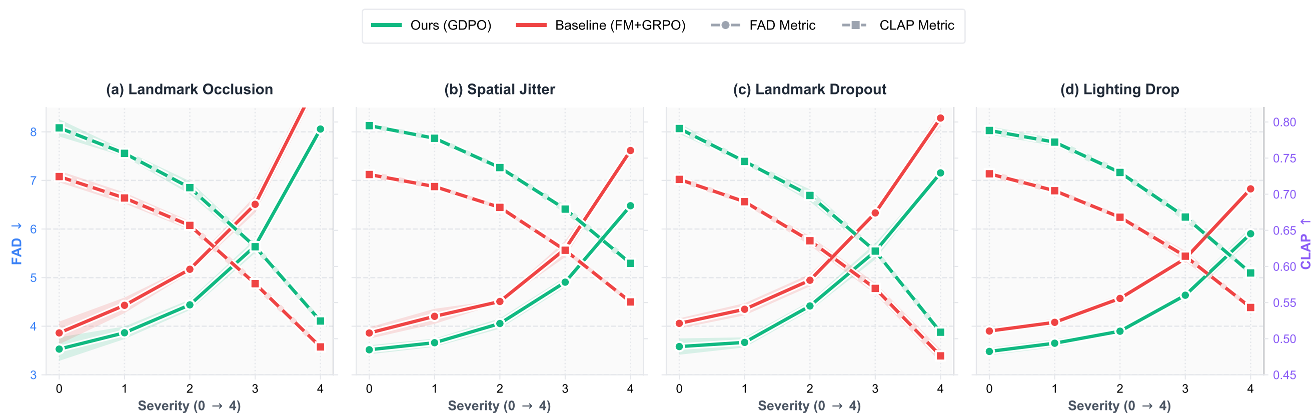

Figure 9. Robustness under visual degradation. FAD (left y-axis, solid lines) and CLAP (right y-axis, dashed lines) for Ours (green) vs. FM+GRPO (red) across four corruption types; $\pm 1\sigma$ shading over 5 seeds.

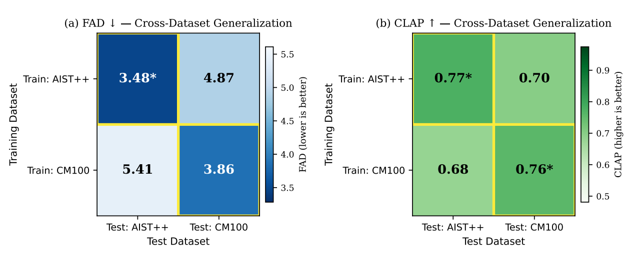

Figure 10. Cross-dataset generalization heatmaps. **(a)** FAD (lower/darker blue = better). **(b)** CLAP (higher/darker green = better). Yellow boxes = in-domain cells.

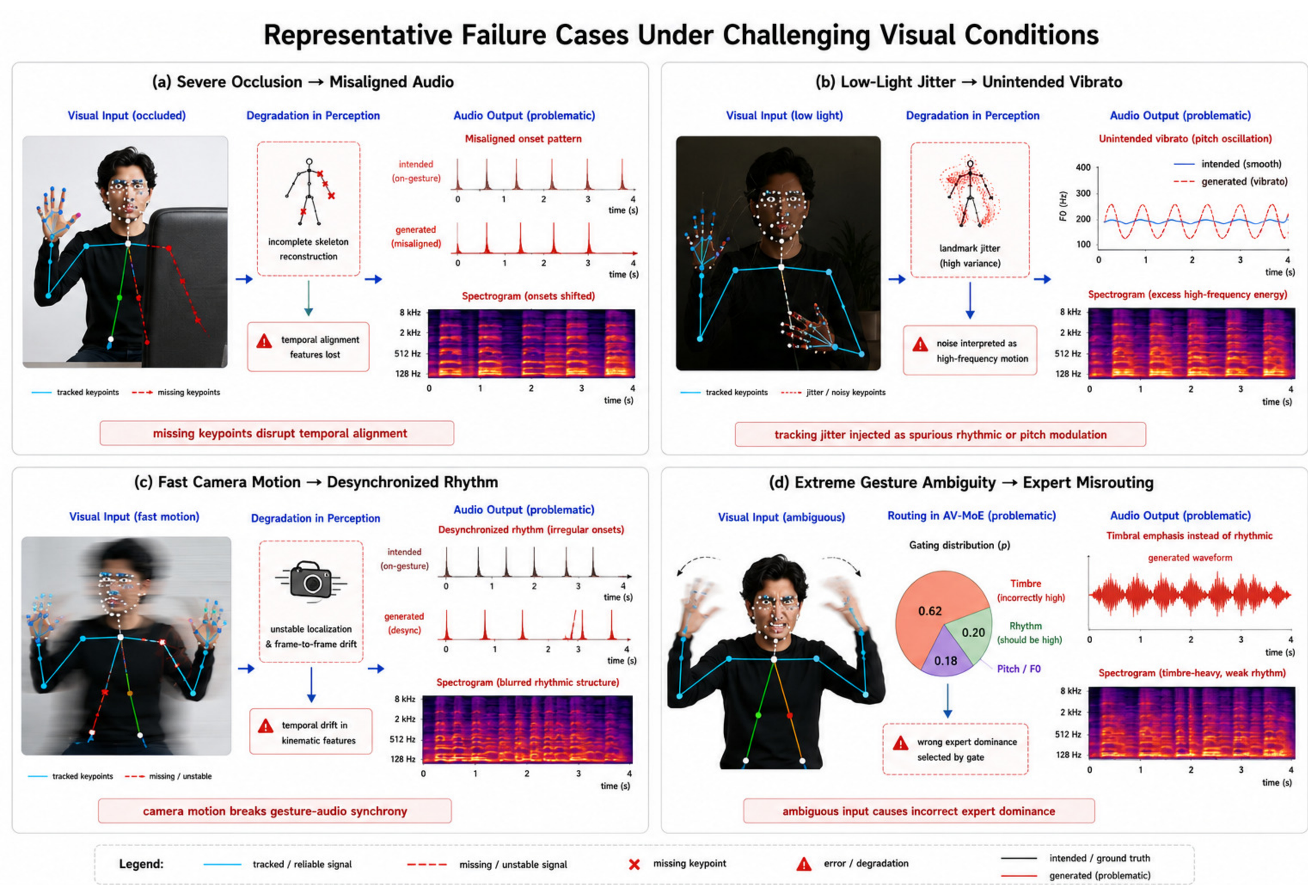

Figure 11. Representative failure cases under challenging visual conditions. **(a)** Severe occlusion leads to incomplete skeletal reconstruction, causing misaligned audio onsets. **(b)** Low-light conditions induce landmark jitter, which the system misinterprets as high-frequency motion, resulting in unintended vibrato. **(c)** Fast camera motion causes frame-to-frame drift, desynchronizing the generated rhythm. **(d)** Extreme gesture ambiguity causes incorrect expert routing, emphasizing timbre over rhythm.

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